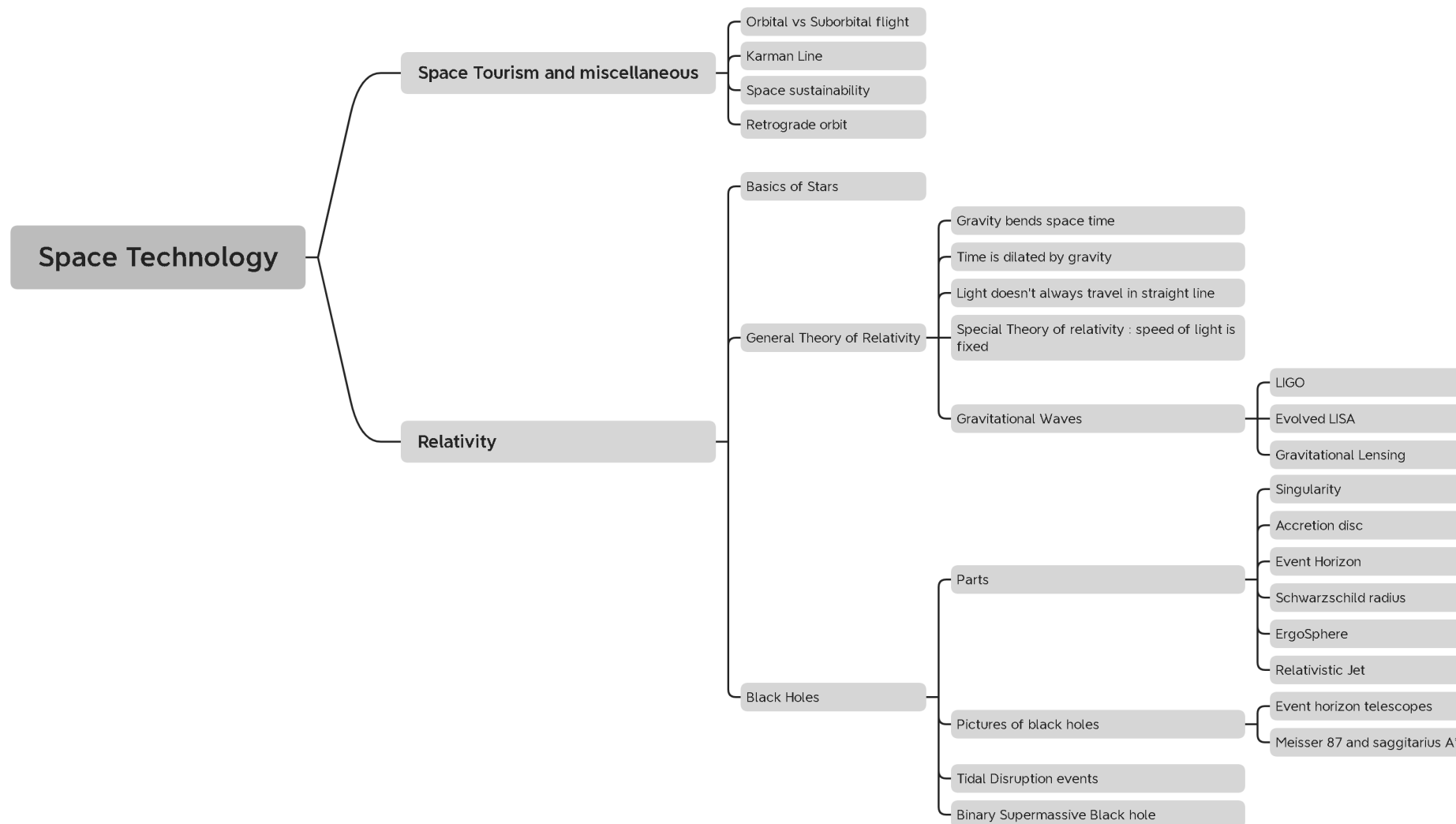
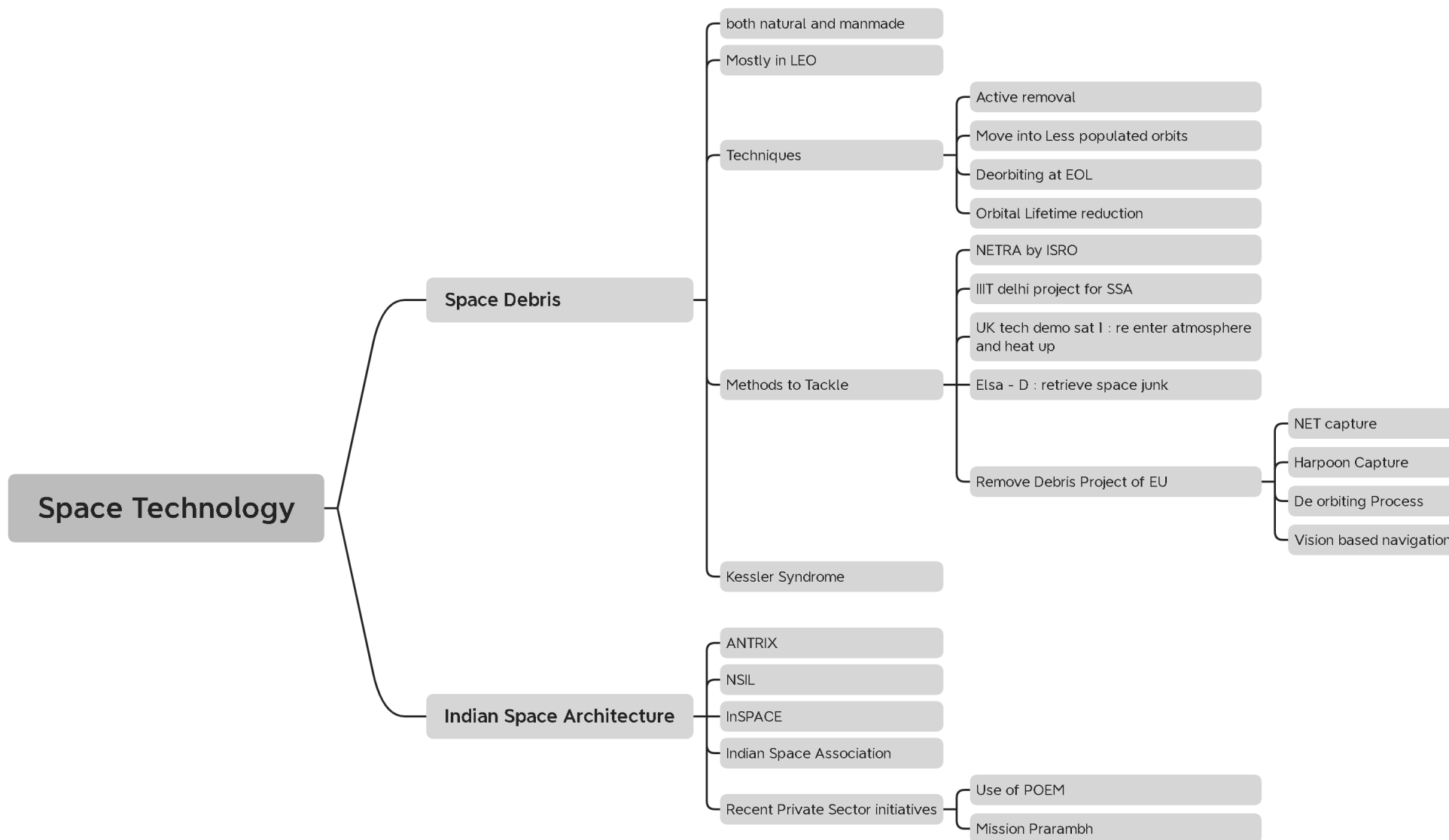
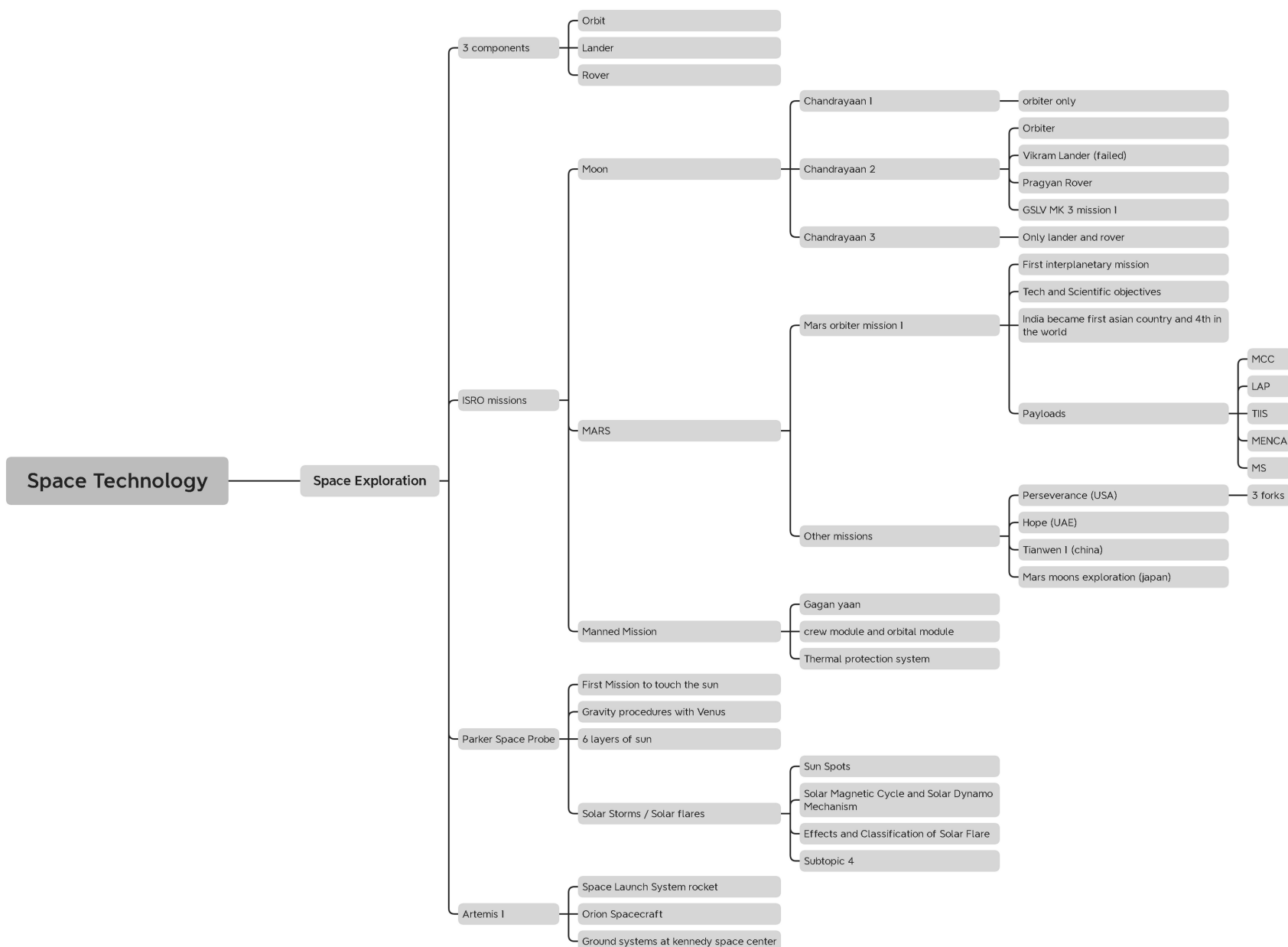
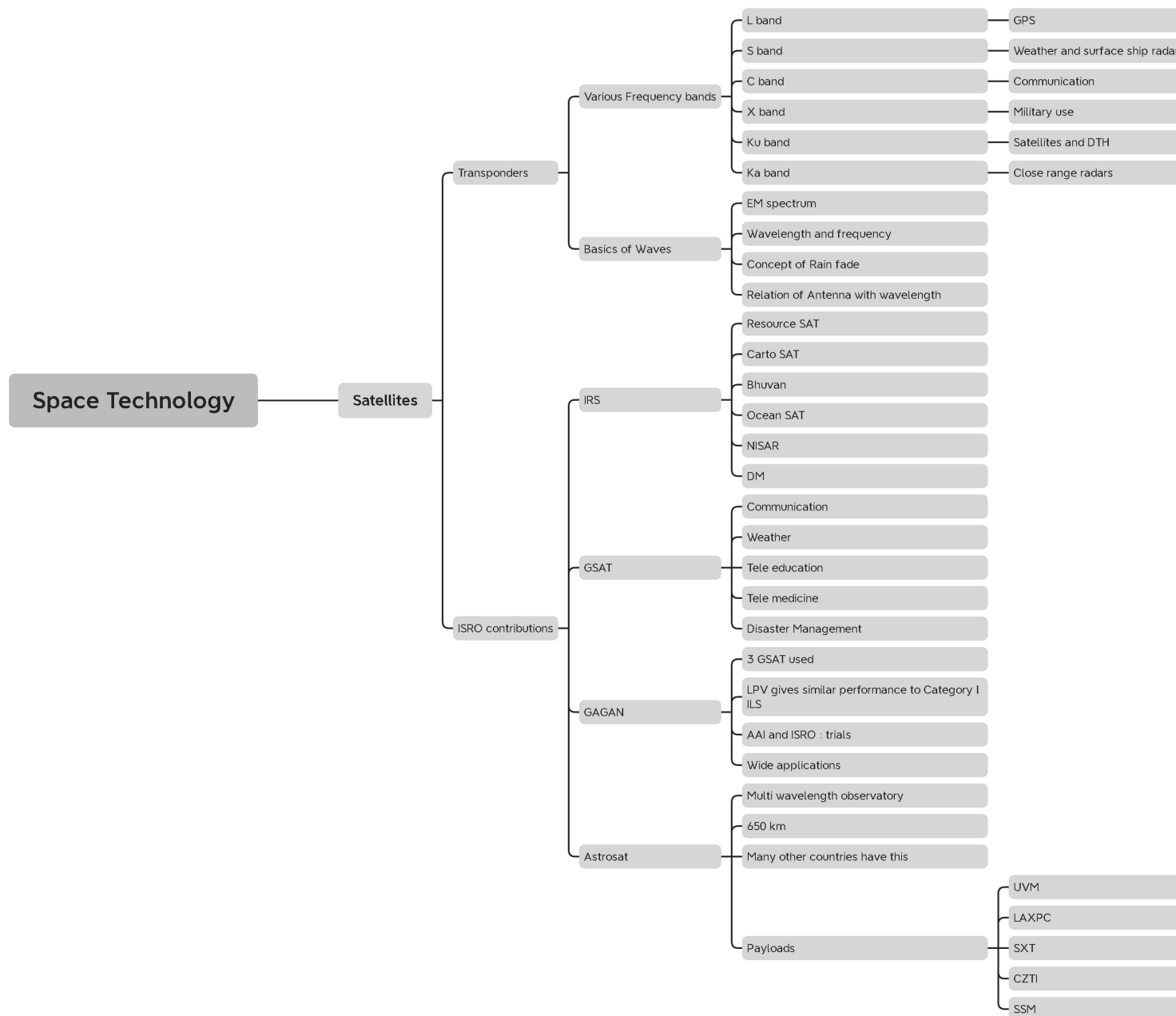












Space Technology

Launch Vehicles

Expendable and Reusable Launch Vehicle

India's Launch vehicles (refer table in slides)

PSLV

GSLV Mark 2

GSLV Mark 3

SSLV

Development of Poem

Engines

Non air breathing

Air breathing

Ramjet : optimal speed mach 3 to mach 6

Scramjet : optimal speed mach 4 to mach 8

Dual Mode Ramjet

Solid Fuel Ducted Ramjet : Indo russian collaboration for missiles

Jet Engine : newton's 3rd law

Fuel

Solid Fuel

Maximum energy

Easy Storage

Heavy Weight

Non Uniform Burn

Liquid Fuel

Good efficiency

Less weight

Difficult storage : risk of leaks

Cryogenic

Liquid hydrogen and oxygen

Environment friendly

